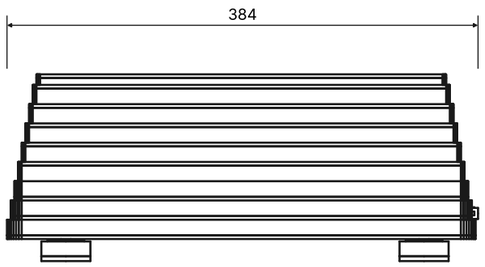
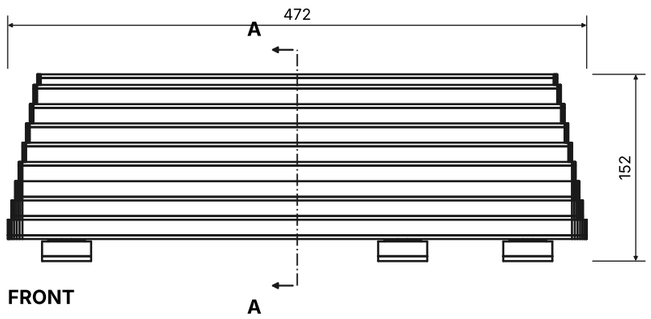
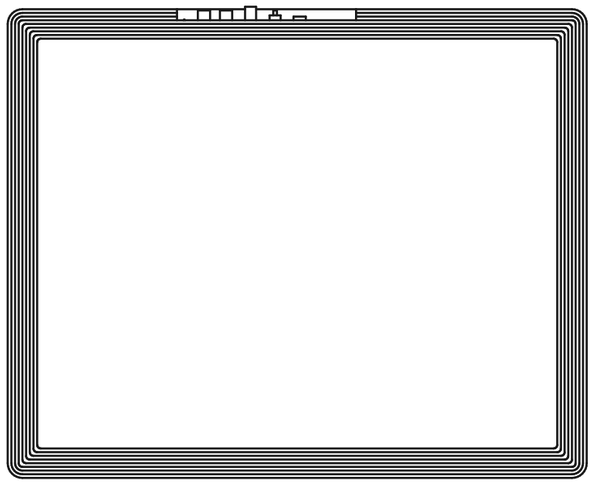
Scale 1:5. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

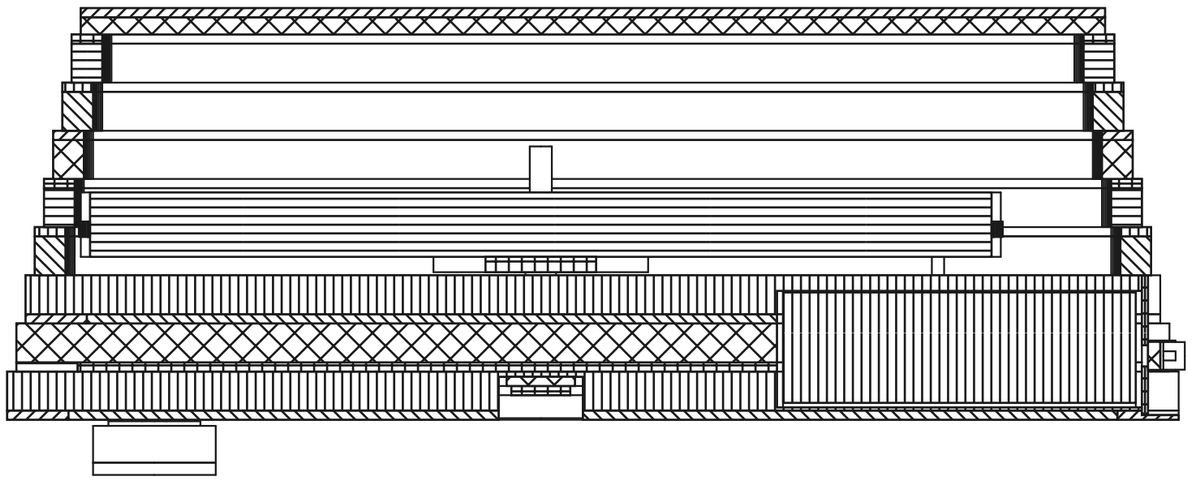
General arrangement

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. Section A-A (sheet 1): the product cut at x 236.0 and seen from the right; cut faces hatched, a different angle for each part.



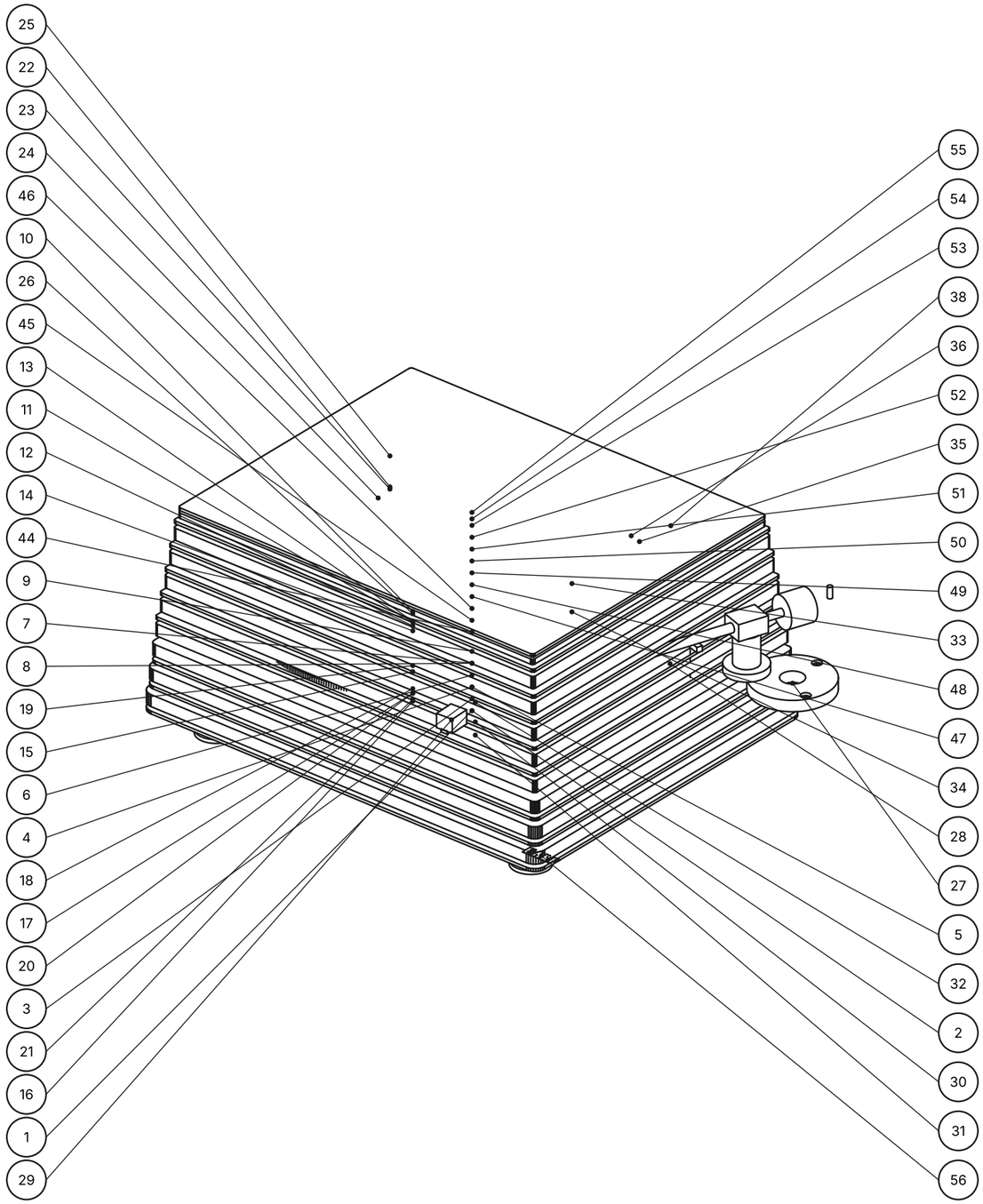
Section A-A

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5, parts drawn apart. Balloons are the parts list's numbers (last sheet).



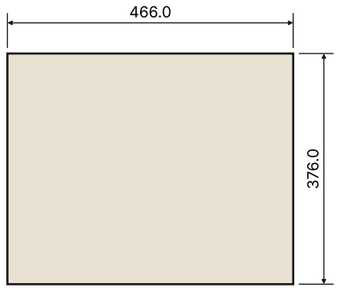
Exploded view

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

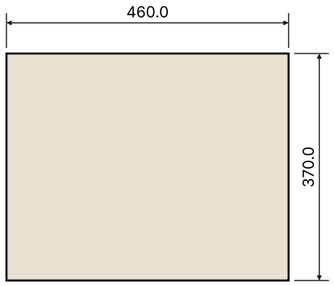
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



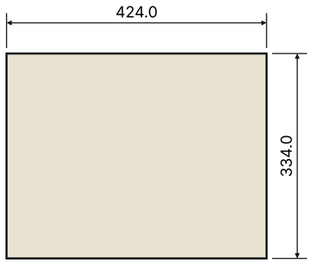
deck-L3

3 mm trucast-tint, 1 for the set



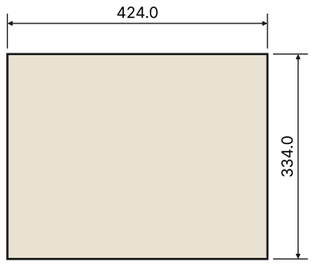
deck-L5

3 mm trucast-tint, 1 for the set



cover-top

5.6 mm trucast-clear, 1 for the set



cover-top-tint

3 mm trucast-tint, 1 for the set

Sheet parts: plain panels

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

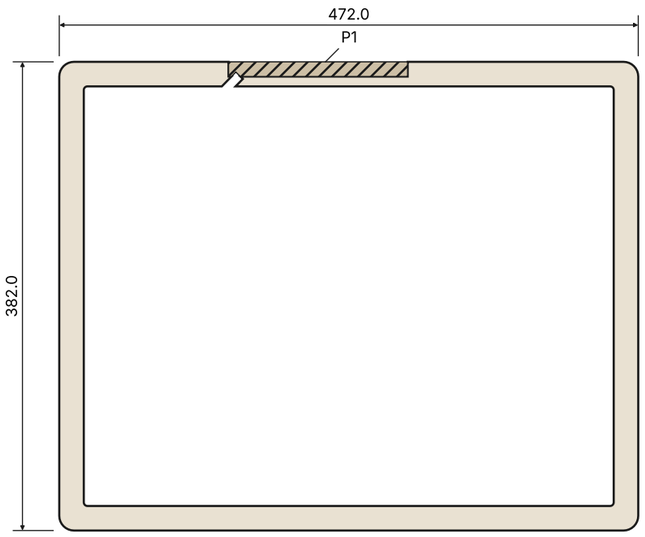
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/deck-L1.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	profile (DXF)	211.0	376.0	1.5 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



deck-L1: 3 mm trucast-tint, 1 for the set, seen from its outer face

not in the cut file: 1 opening(s) in the inner face that reach neither face nor a pocket

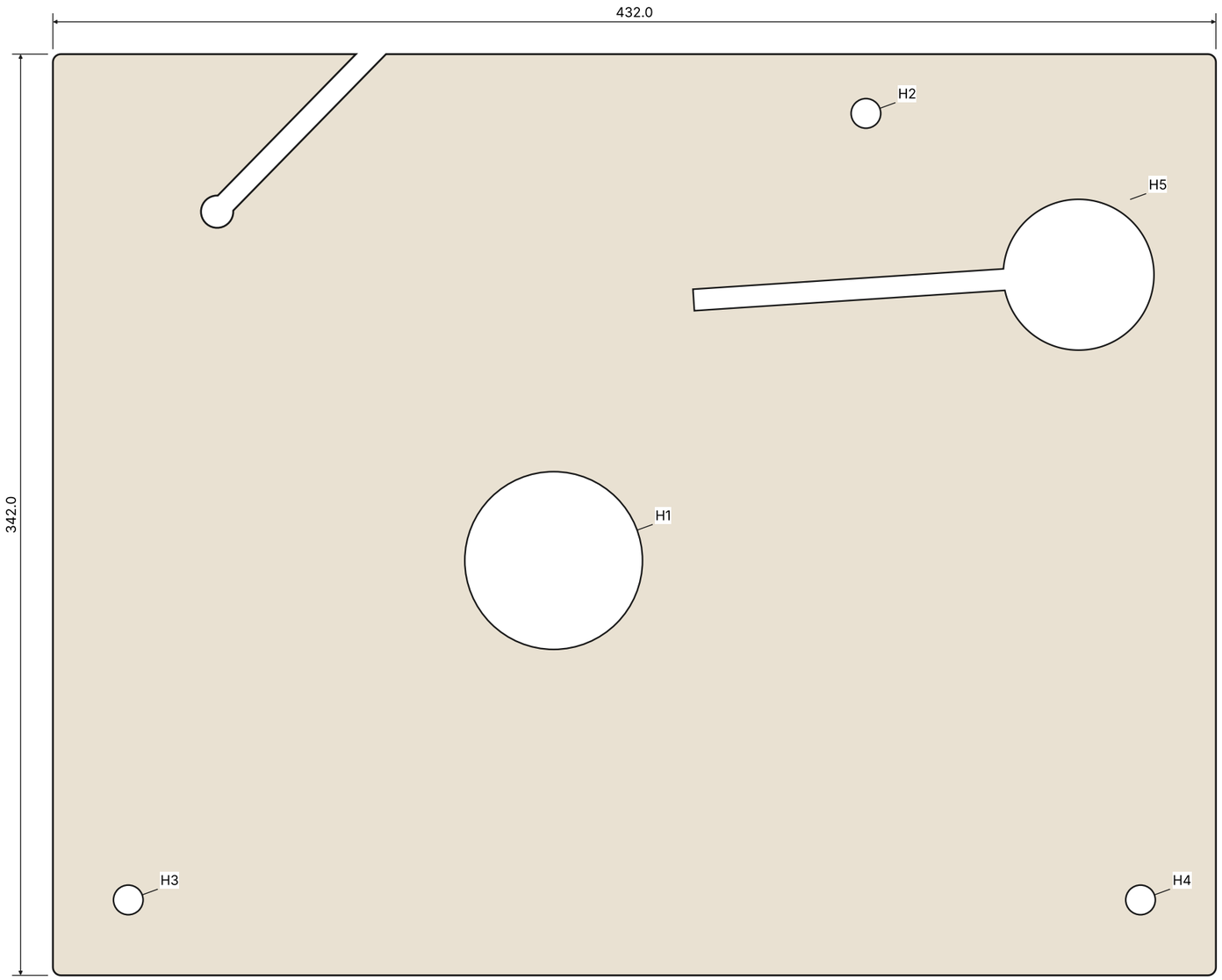
Sheet part: deck-L1

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/deck-L1-inlay.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	Ø66.0	186.0	154.0	through
H2	Ø11.0	302.0	320.0	through
H3	Ø11.0	28.0	28.0	through
H4	Ø11.0	404.0	28.0	through
H5	profile (DXF)	323.4	260.1	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

deck-L1-inlay: 3 mm trucast-clear, 1 for the set, seen from its outer face

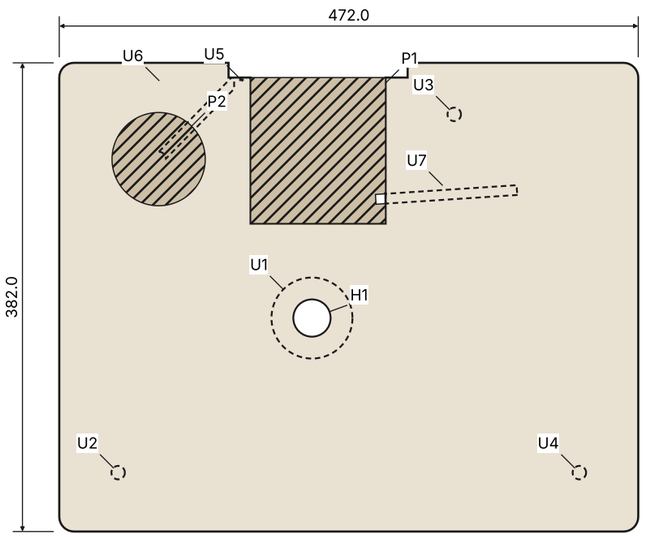
Sheet part: deck-L1-inlay

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/deck-L2.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	110.0 x 119.0	211.0	310.5	11.7 from face
P2	Ø76.0	81.0	303.5	8.6 from face
H1	Ø30.4	206.0	174.0	through
U1	Ø66.0	206.0	174.0	11 from inside
U2	Ø11.0	48.0	48.0	10 from inside
U3	Ø11.0	322.0	340.0	10 from inside
U4	Ø11.0	424.0	48.0	10 from inside
U5	profile (DXF)	148.6	367.9	3 from inside
U6	profile (DXF)	111.9	337.0	3 from inside
U7	profile (DXF)	319.8	274.8	3 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

deck-L2: 12.7 mm trucast-clear, 1 for the set, seen from its outer face

inner-face pockets: dxf/deck-L2-underside.dxf (seen from that face)

not in the cut file: 1 opening(s) in the inner face that reach neither face nor a pocket

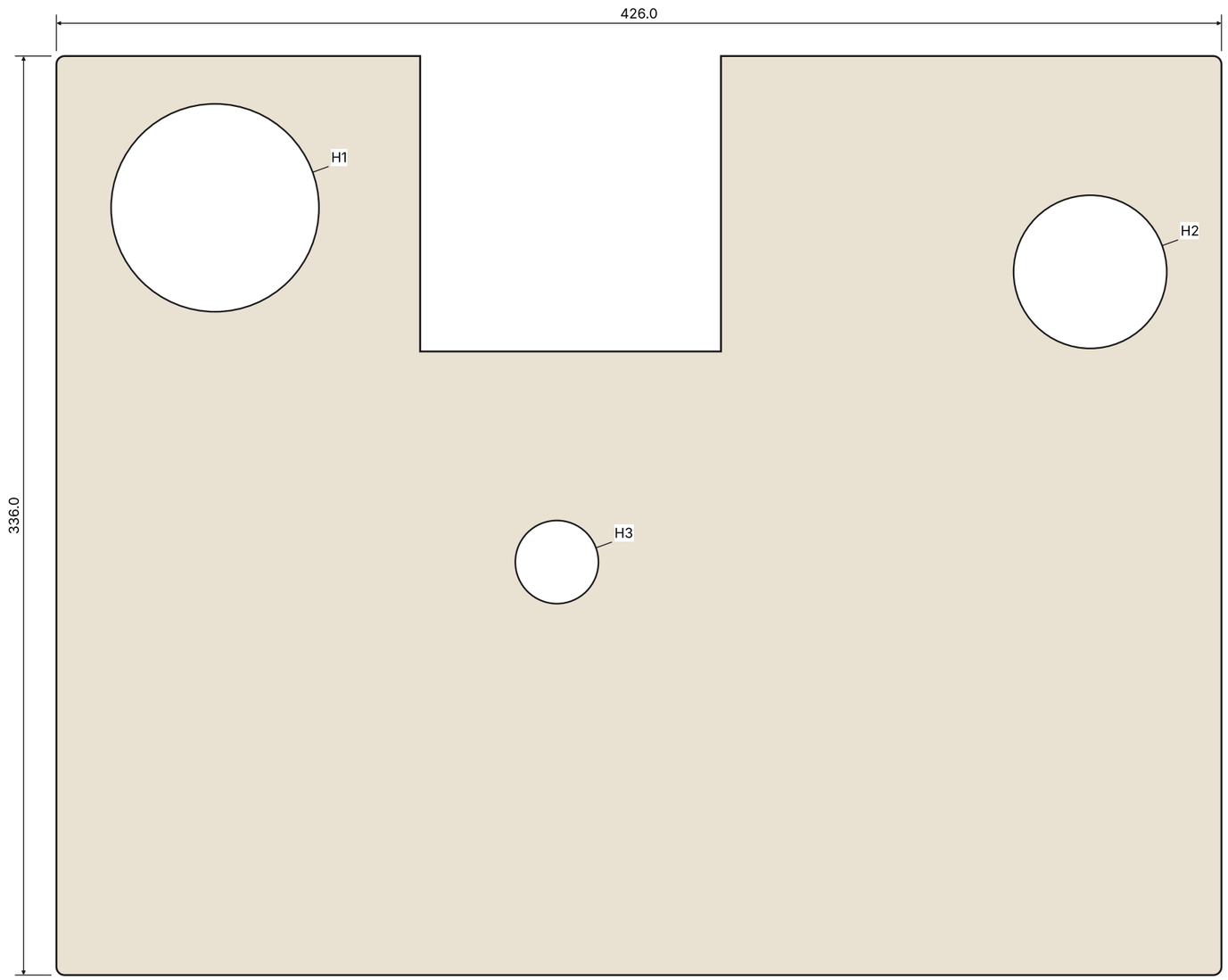
Sheet part: deck-L2

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/deck-L3-inlay.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	Ø76.0	58.0	280.5	through
H2	Ø56.0	378.0	257.1	through
H3	Ø30.4	183.0	151.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

deck-L3-inlay: 3 mm trucast-clear, 1 for the set, seen from its outer face

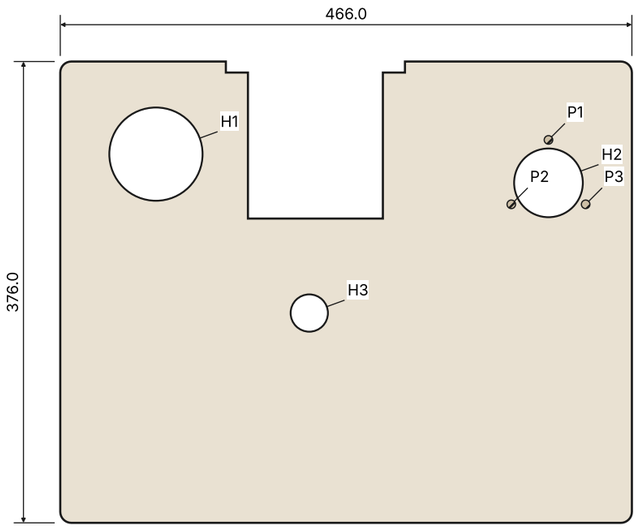
Sheet part: deck-L3-inlay

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/deck-L4.dxf. Marks listed at the right.



deck-L4: 12.7 mm trucast-clear, 1 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø7.0	398.0	312.1	10.3 from face
P2	Ø7.0	367.7	259.6	10.3 from face
P3	Ø7.0	428.3	259.6	10.3 from face
H1	Ø76.0	78.0	300.5	through
H2	Ø56.0	398.0	277.1	through
H3	Ø30.4	203.0	171.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

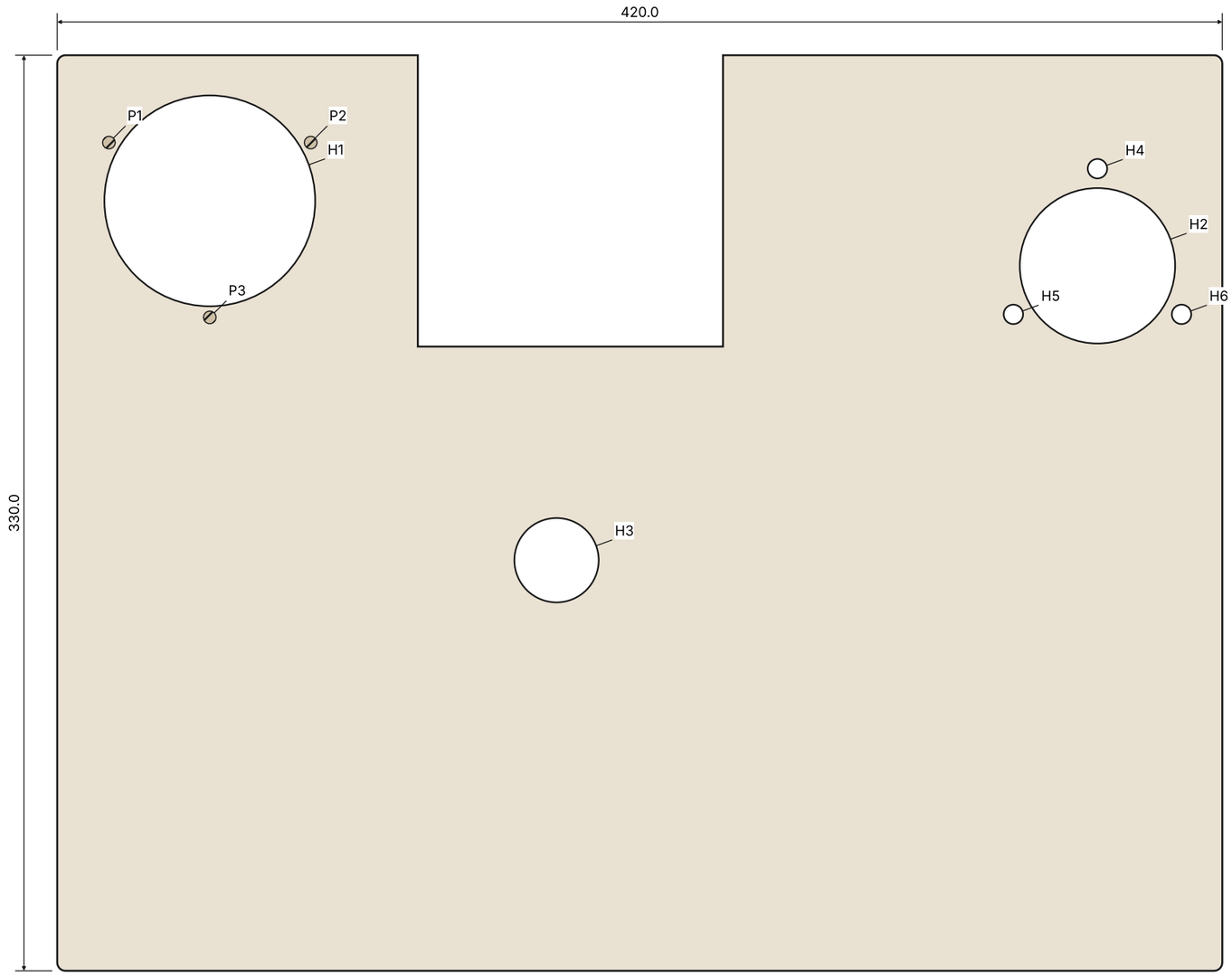
Sheet part: deck-L4

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/deck-L5-inlay.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	Ø4.6	18.6	298.5	1.3 from face
P2	Ø4.6	91.4	298.5	1.3 from face
P3	Ø4.6	55.0	235.5	1.3 from face
H1	Ø76.0	55.0	277.5	through
H2	Ø56.0	375.0	254.1	through
H3	Ø30.4	180.0	148.0	through
H4	Ø7.0	375.0	289.1	through
H5	Ø7.0	344.7	236.6	through
H6	Ø7.0	405.3	236.6	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

deck-L5-inlay: 3 mm trucast-clear, 1 for the set, seen from its outer face

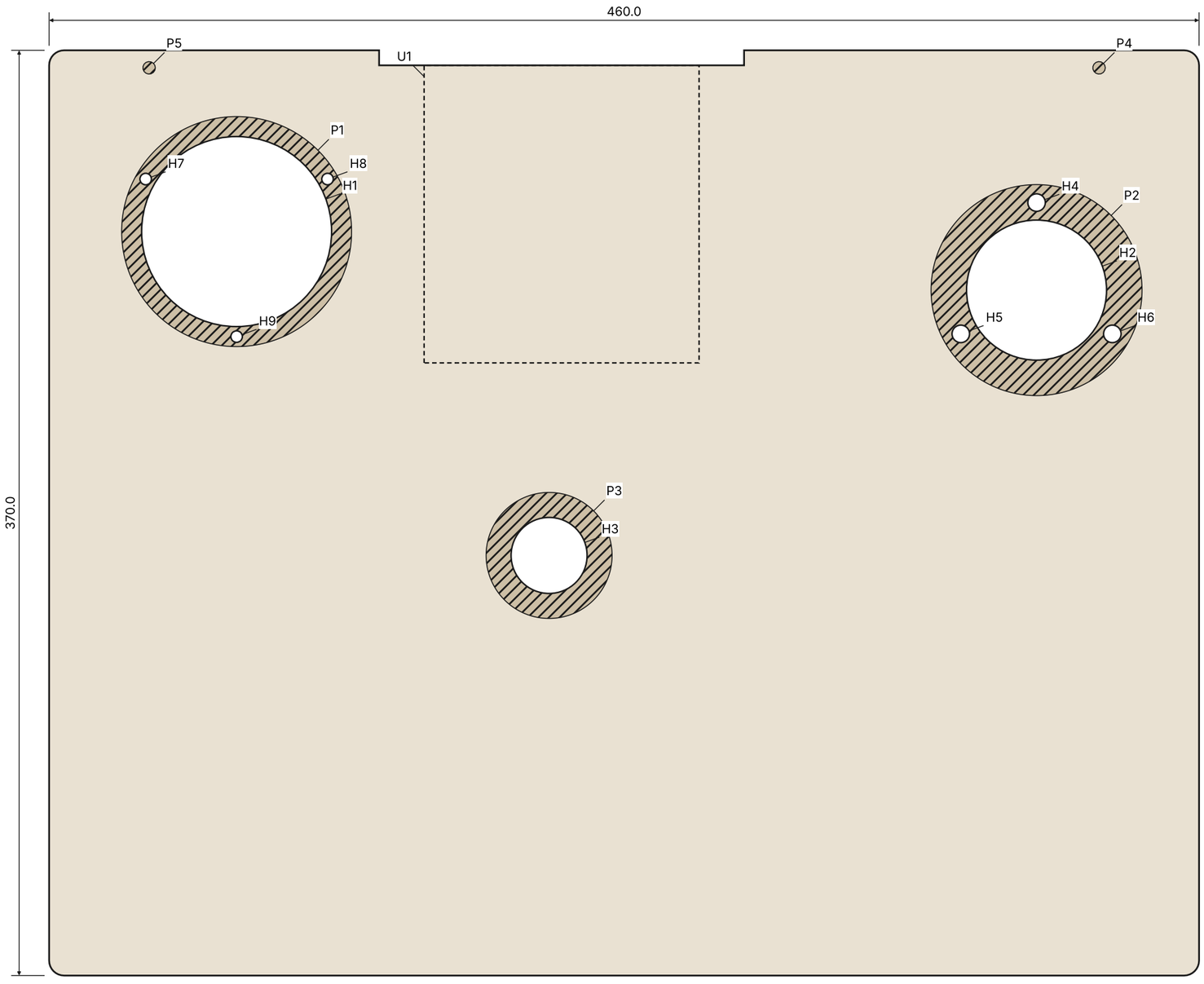
Sheet part: deck-L5-inlay

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/deck-L6.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	Ø92.0	75.0	297.5	6 from face
P2	Ø84.4	395.0	274.1	12 from face
P3	Ø50.4	200.0	168.0	5 from face
P4	Ø5.0	420.0	363.0	8 from face
P5	Ø5.0	40.0	363.0	8 from face
H1	Ø76.0	75.0	297.5	through
H2	Ø56.0	395.0	274.1	through
H3	Ø30.4	200.0	168.0	through
H4	Ø7.0	395.0	309.1	through
H5	Ø7.0	364.7	256.6	through
H6	Ø7.0	425.3	256.6	through
H7	Ø4.6	38.6	318.5	through
H8	Ø4.6	111.4	318.5	through
H9	Ø4.6	75.0	255.5	through
U1	110.0 x 119.0	205.0	304.5	7.6 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

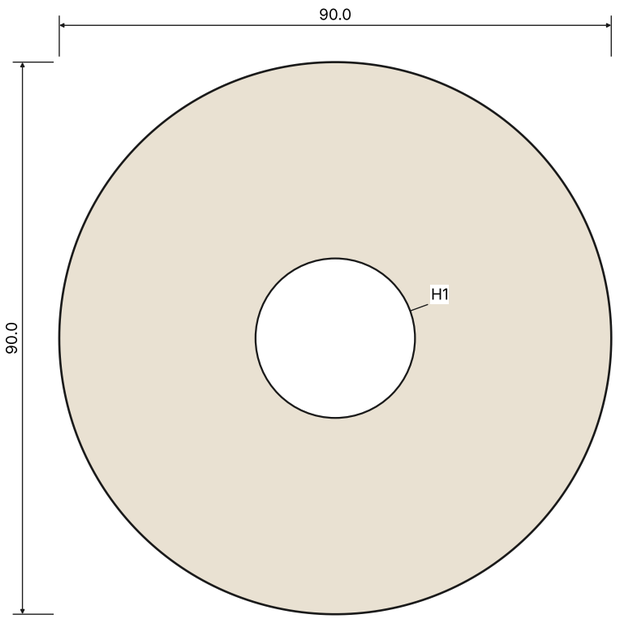
deck-L6: 12.7 mm trucast-clear, 1 for the set, seen from its outer face
inner-face pockets: dxf/deck-L6-underside.dxf (seen from that face)

Sheet part: deck-L6

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/motor-plate.dxf. Marks listed at the right.



motor-plate: 3 mm aluminium, 1 for the set, seen from its outer face

mark	size	x	y	depth
H1	Ø26.0	45.0	45.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: motor-plate

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

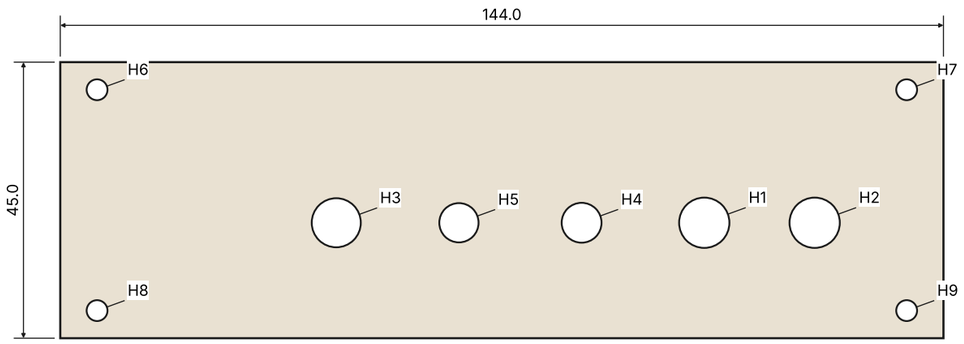
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/back-plate.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø8.2	105.0	18.8	through
H2	Ø8.2	123.0	18.8	through
H3	Ø8.0	45.0	18.8	through
H4	Ø6.5	85.0	18.8	through
H5	Ø6.4	65.0	18.8	through
H6	Ø3.4	6.0	40.5	through
H7	Ø3.4	138.0	40.5	through
H8	Ø3.4	6.0	4.5	through
H9	Ø3.4	138.0	4.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



back-plate: 2 mm brass-sheet, 1 for the set, seen from its outer face

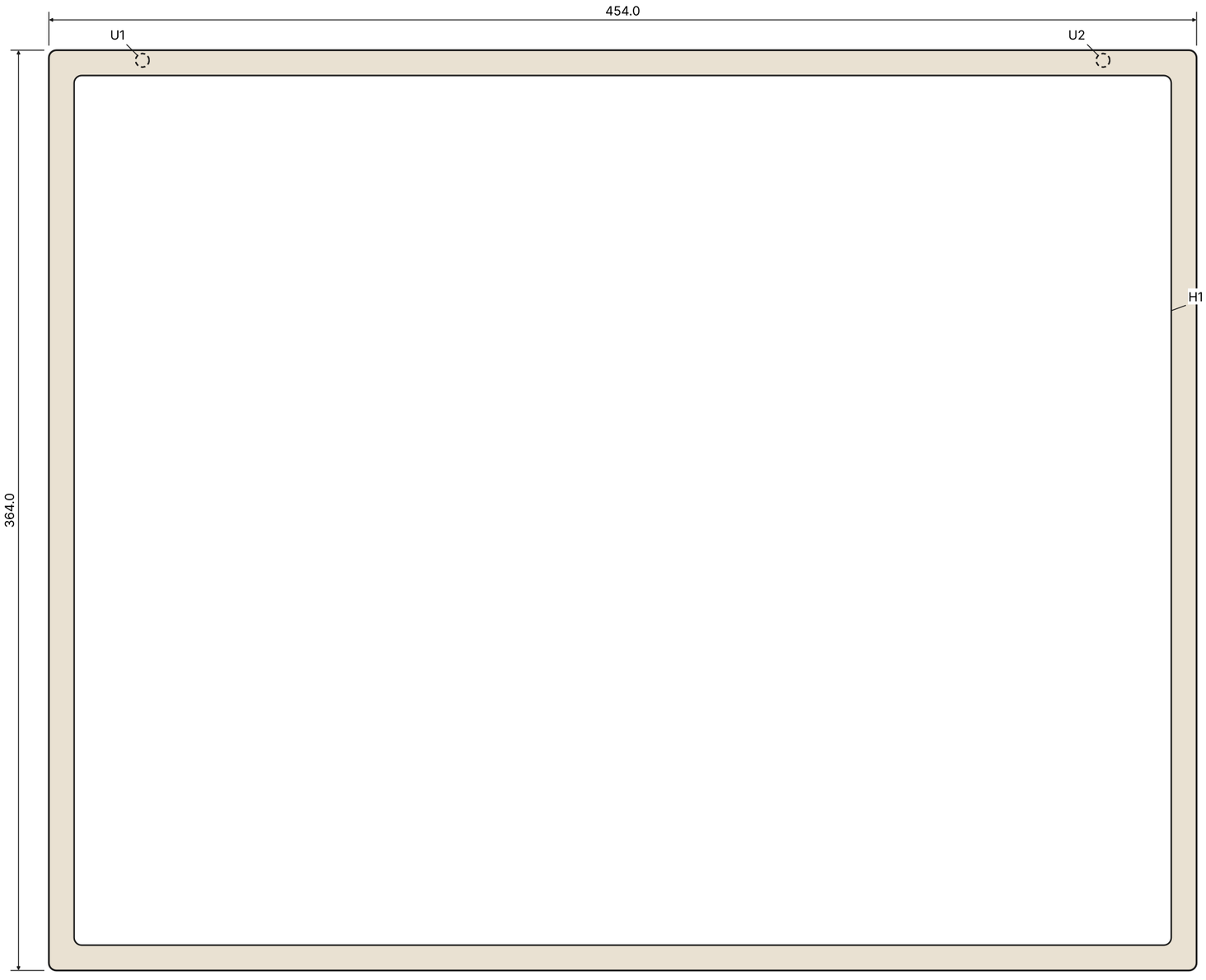
Sheet part: back-plate

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/cover-C1.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	profile (DXF)	227.0	182.0	through
U1	Ø5.4	37.0	360.0	8 from inside
U2	Ø5.4	417.0	360.0	8 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

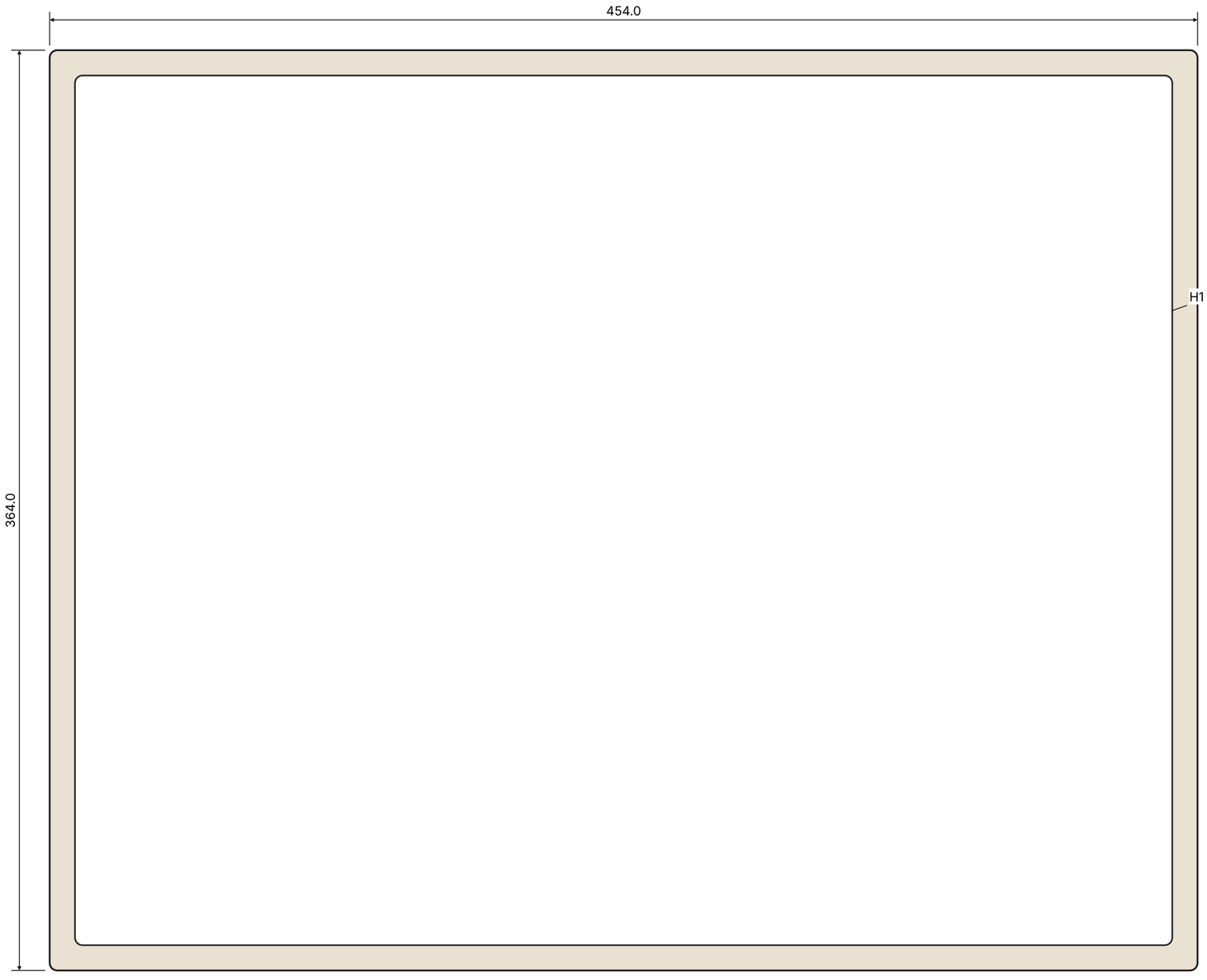
cover-C1: 12.7 mm trucast-clear, 1 for the set, seen from its outer face
inner-face pockets: dxf/cover-C1-underside.dxf (seen from that face)

Sheet part: cover-C1

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/cover-C2.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	profile (DXF)	227.0	182.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

cover-C2: 3 mm trucast-tint, 1 for the set, seen from its outer face

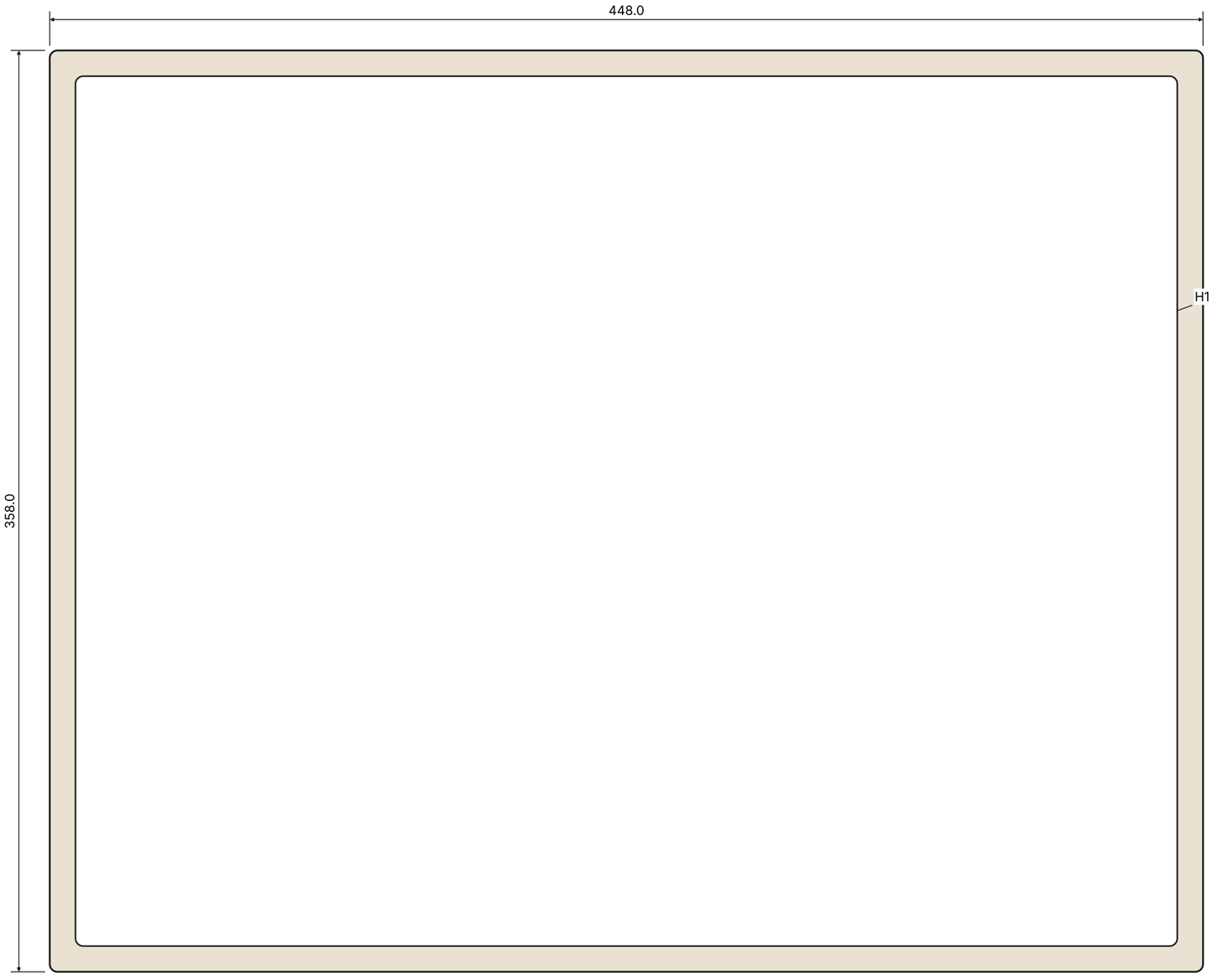
Sheet part: cover-C2

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/cover-C3.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	profile (DXF)	224.0	179.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

cover-C3: 12.7 mm trucast-clear, 1 for the set, seen from its outer face

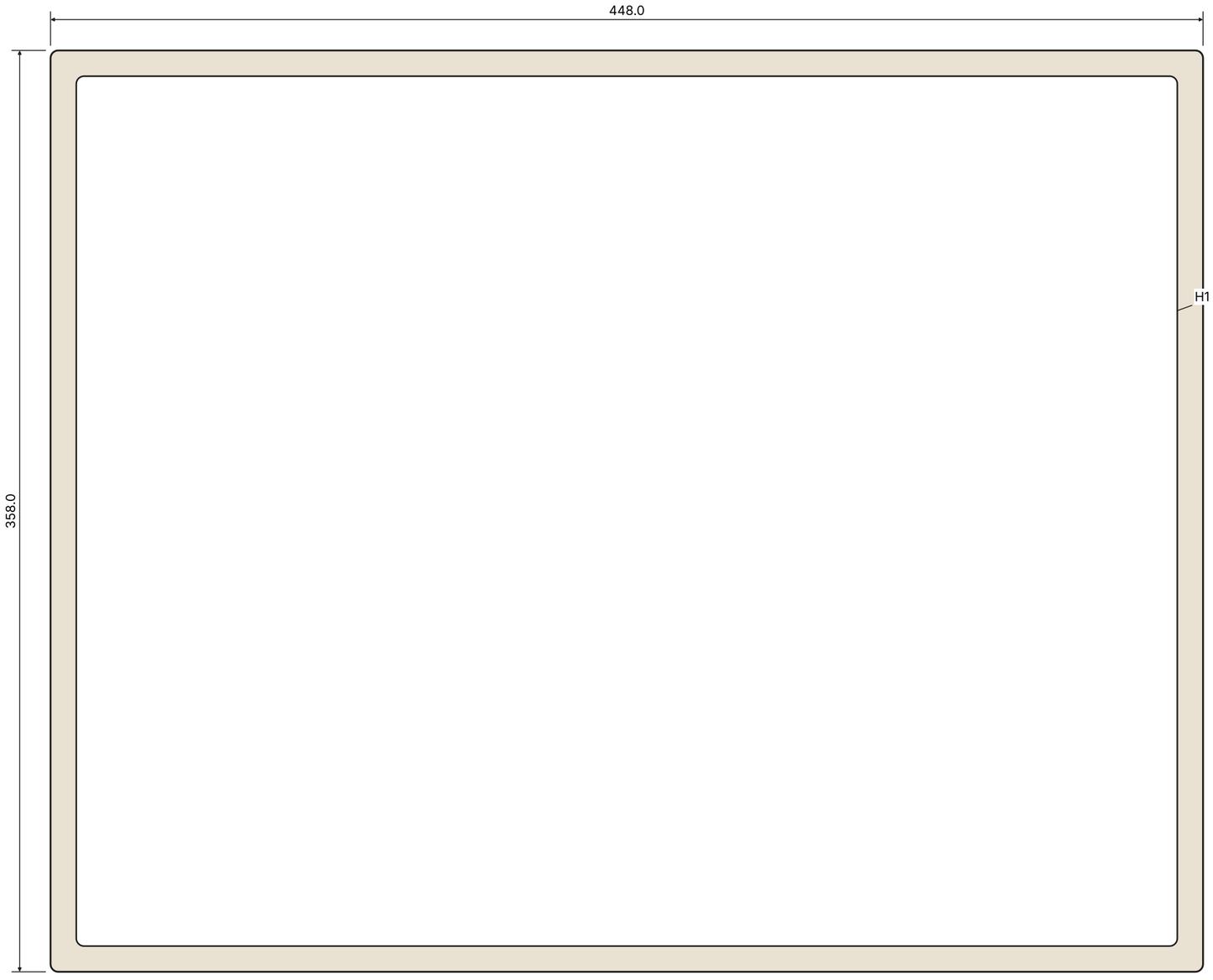
Sheet part: cover-C3

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/cover-C4.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	profile (DXF)	224.0	179.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

cover-C4: 3 mm trucast-tint, 1 for the set, seen from its outer face

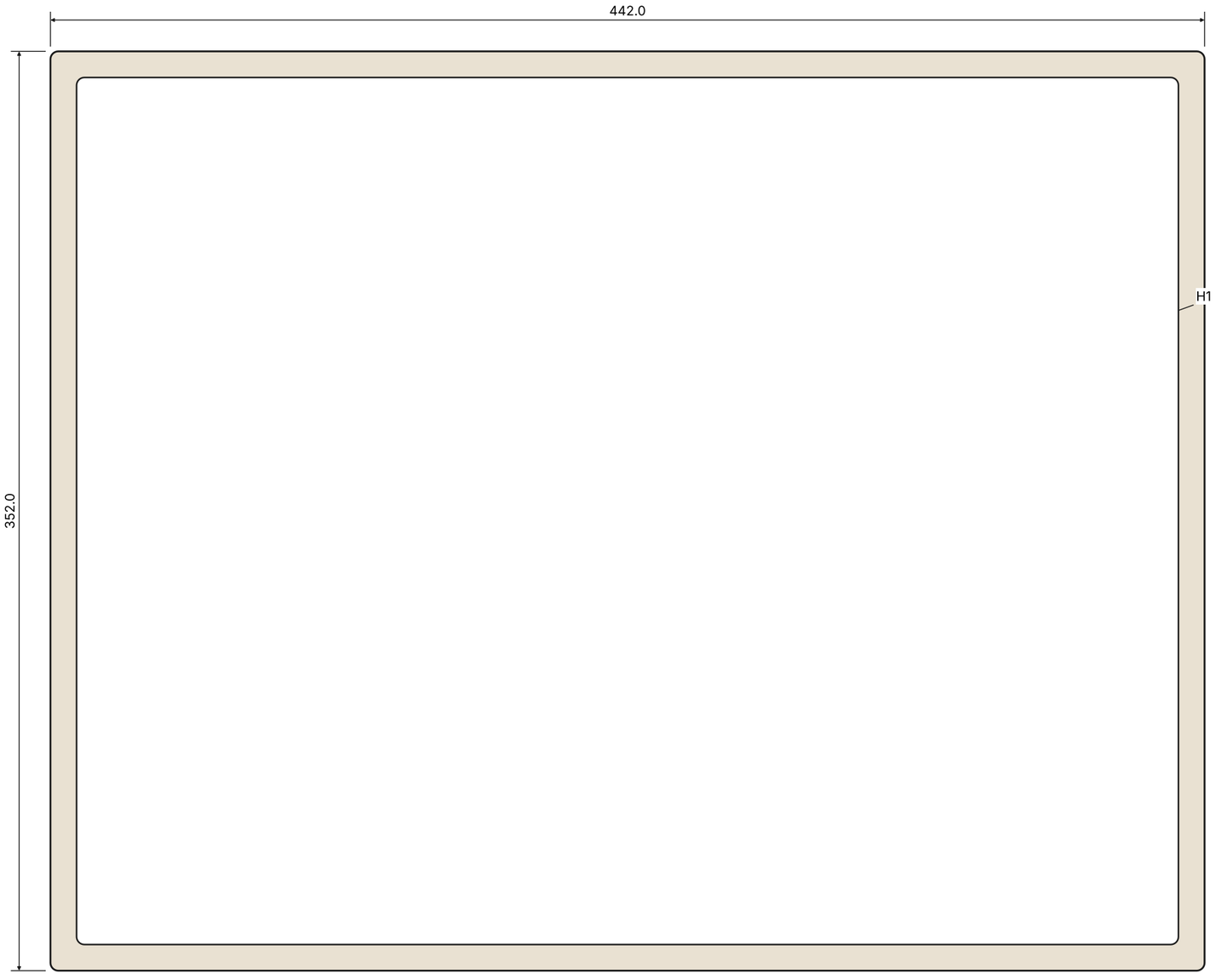
Sheet part: cover-C4

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/cover-C5.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	profile (DXF)	221.0	176.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

cover-C5: 12.7 mm trucast-clear, 1 for the set, seen from its outer face

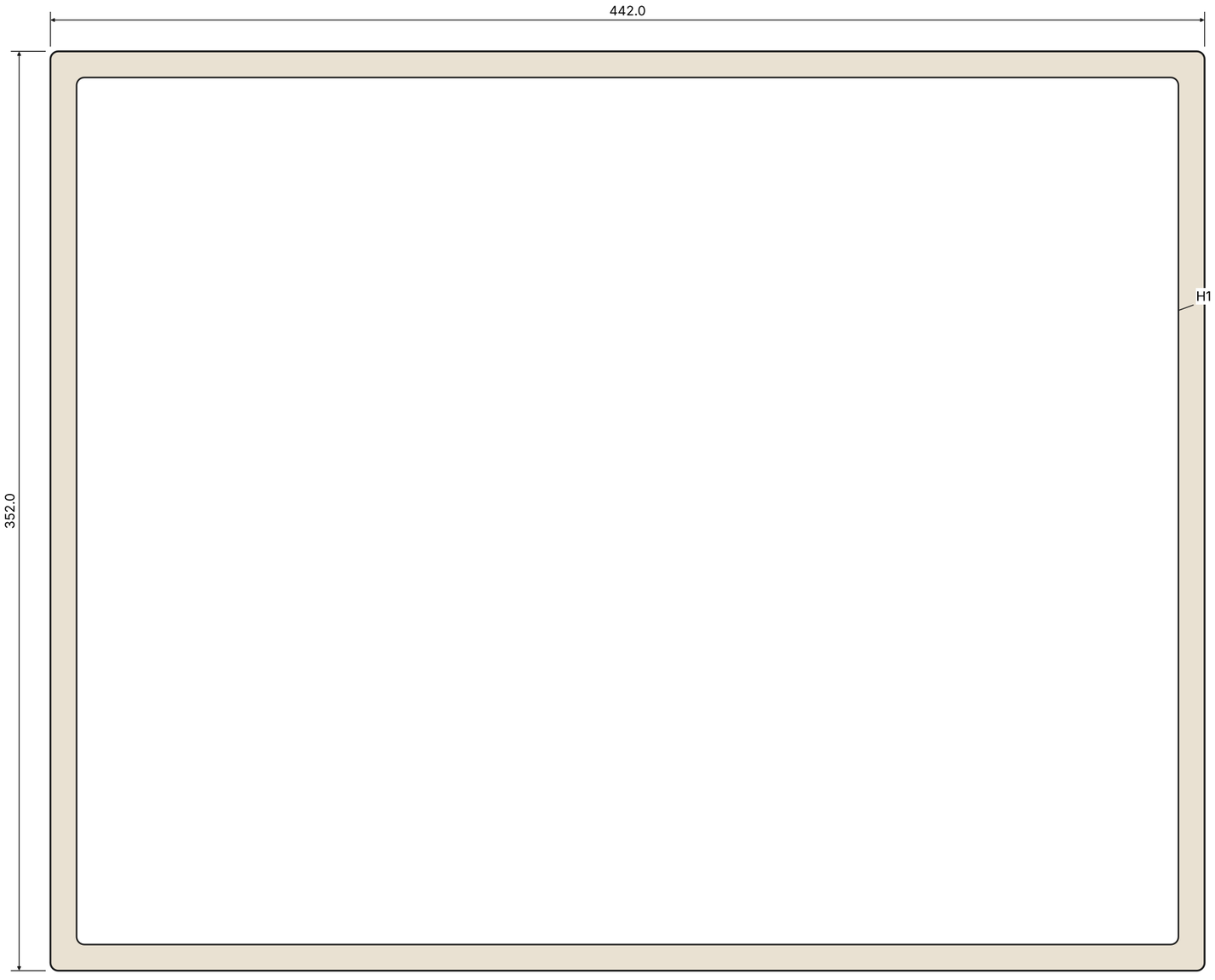
Sheet part: cover-C5

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/cover-C6.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	profile (DXF)	221.0	176.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

cover-C6: 3 mm trucast-tint, 1 for the set, seen from its outer face

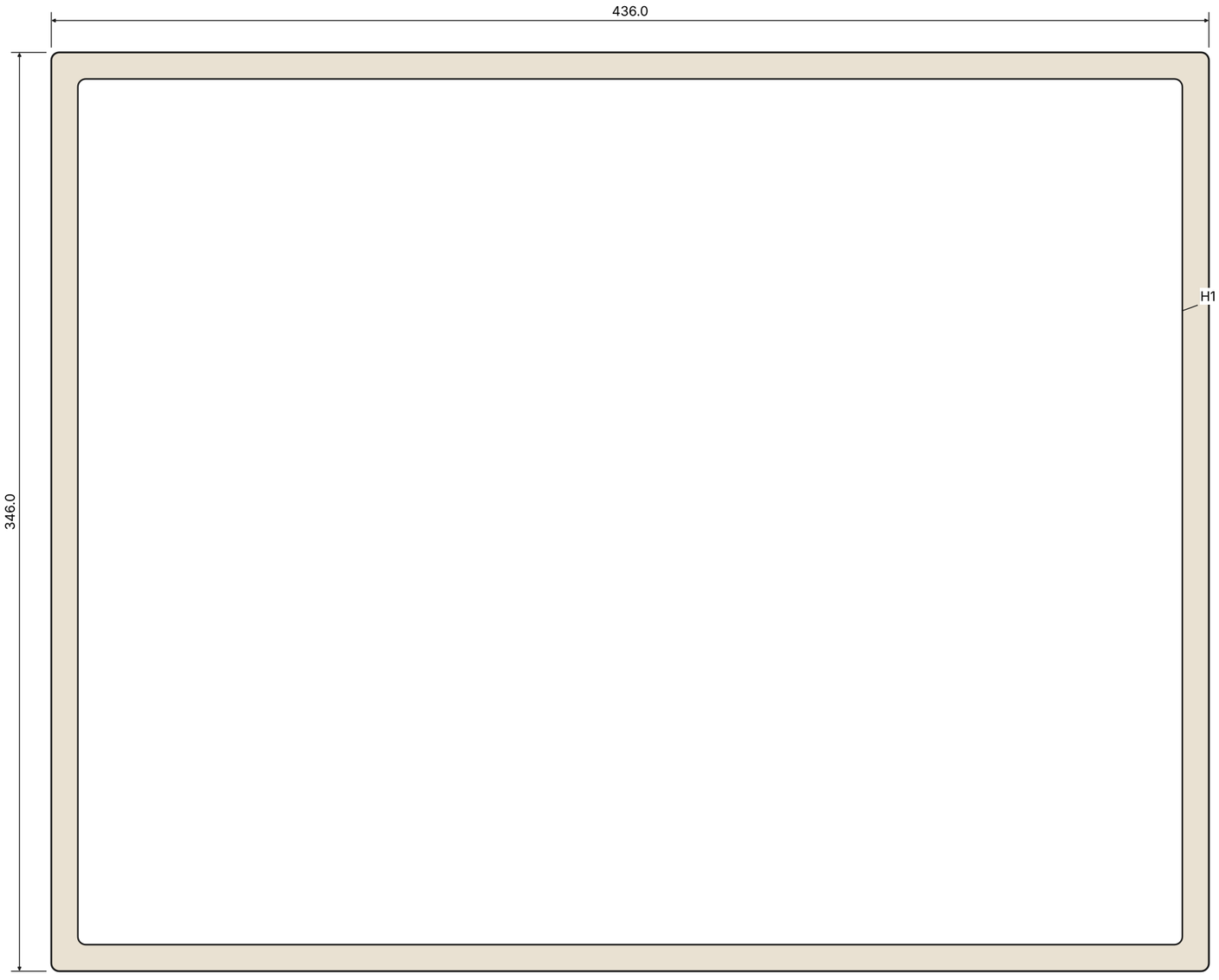
Sheet part: cover-C6

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/cover-C7.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	profile (DXF)	218.0	173.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

cover-C7: 12.7 mm trucast-clear, 1 for the set, seen from its outer face

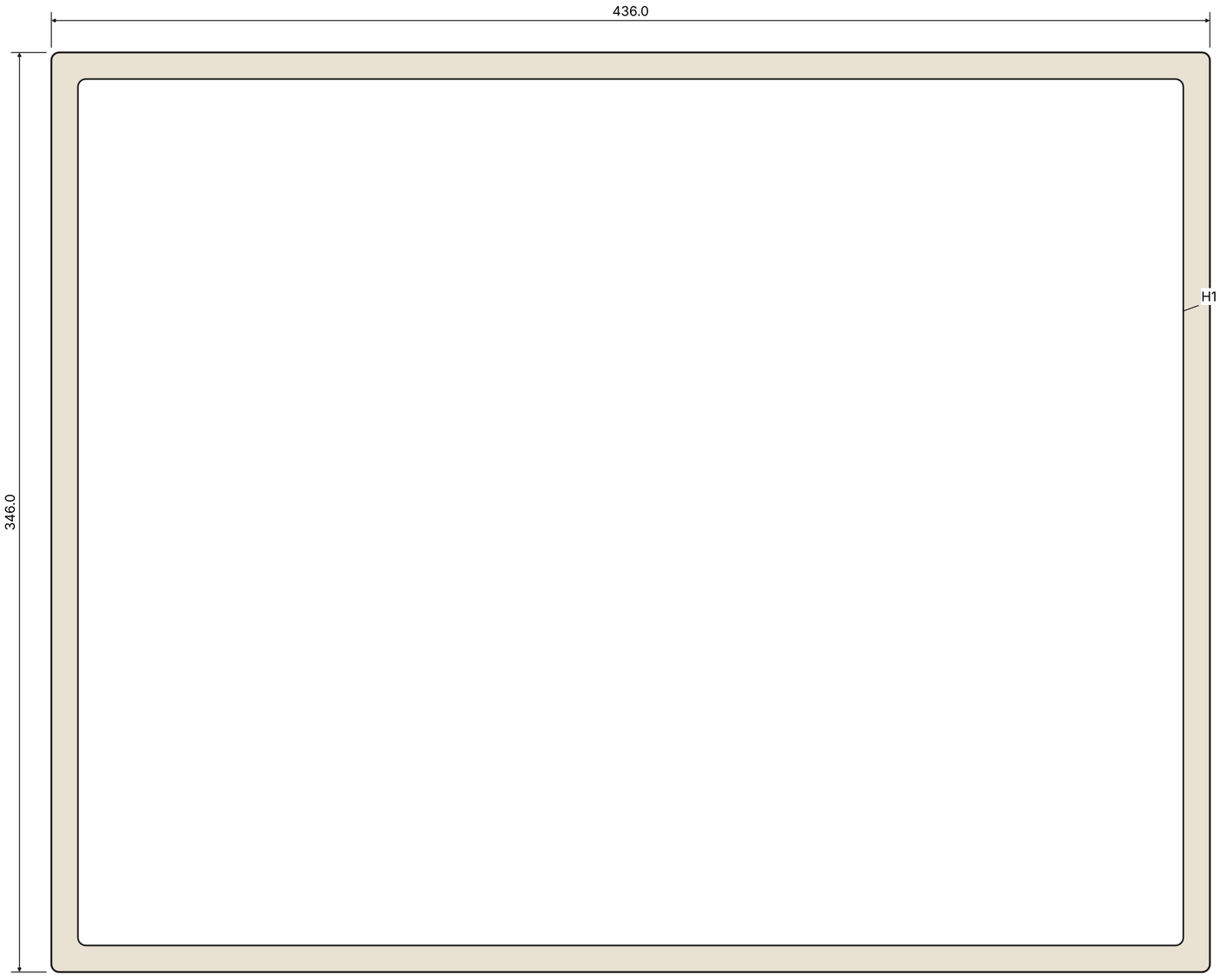
Sheet part: cover-C7

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/cover-C8.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	profile (DXF)	218.0	173.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

cover-C8: 3 mm trucast-tint, 1 for the set, seen from its outer face

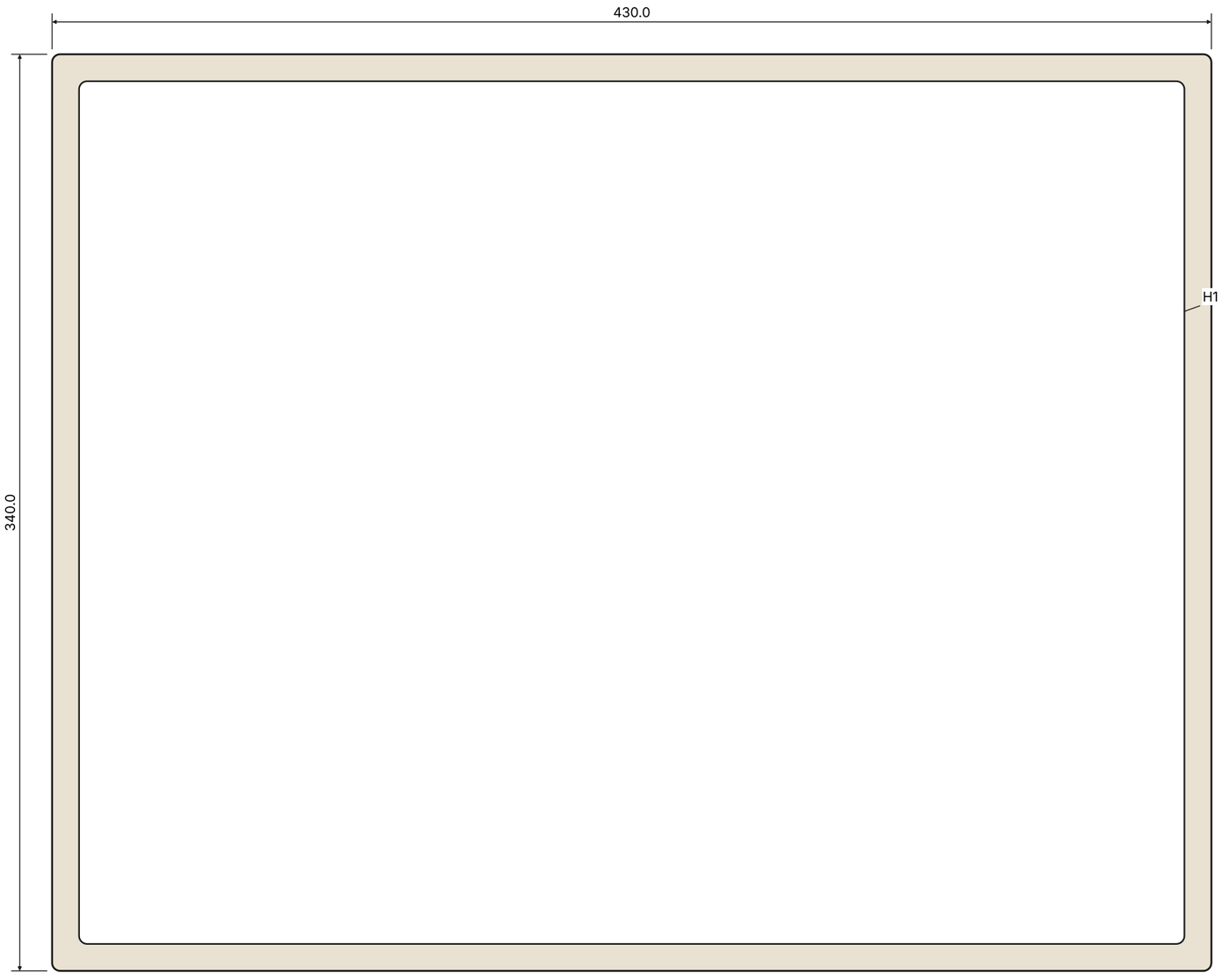
Sheet part: cover-C8

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/cover-C9.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	profile (DXF)	215.0	170.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

cover-C9: 12.7 mm trucast-clear, 1 for the set, seen from its outer face

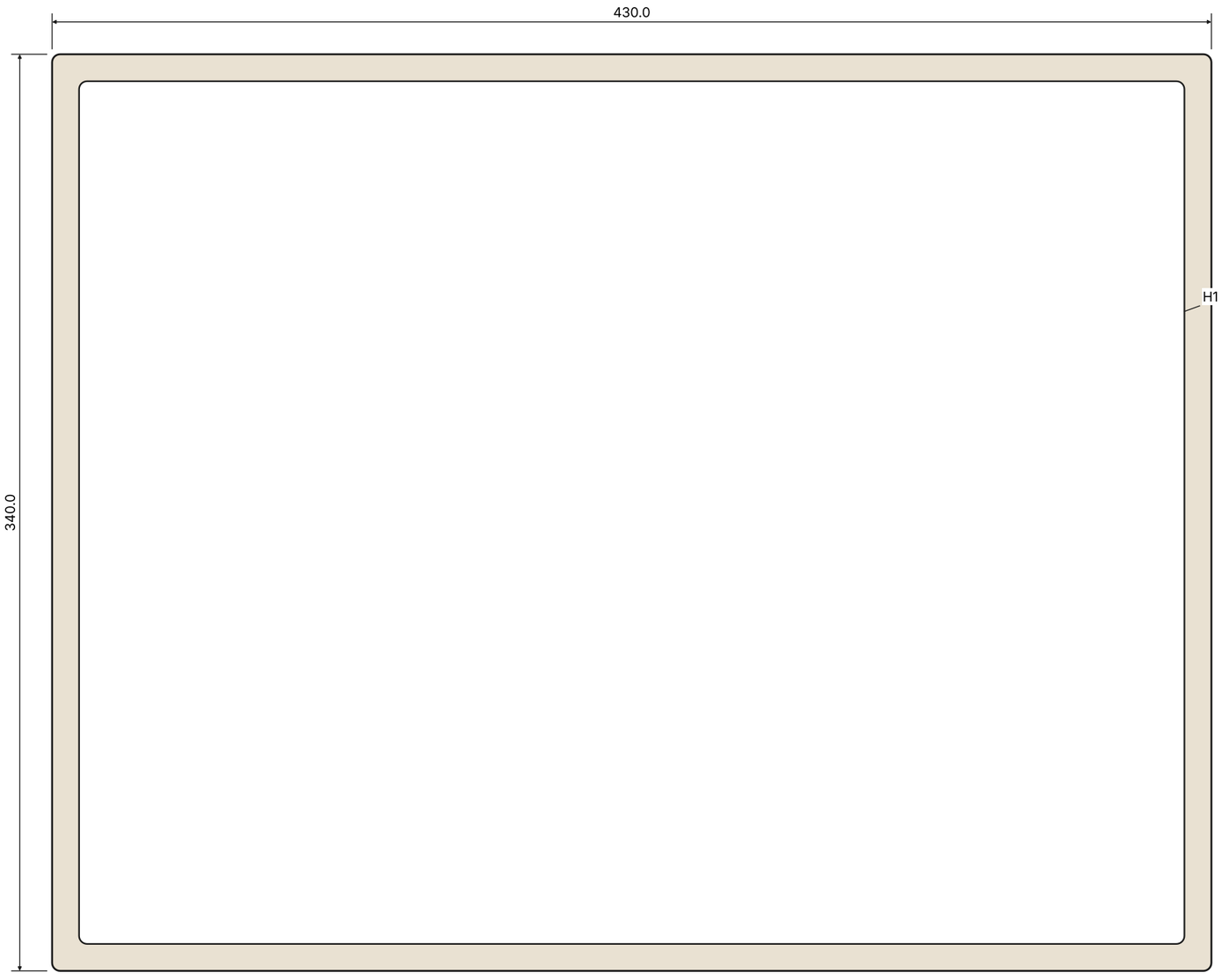
Sheet part: cover-C9

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/cover-C10.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	profile (DXF)	215.0	170.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

cover-C10: 3 mm trucast-tint, 1 for the set, seen from its outer face

Sheet part: cover-C10

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

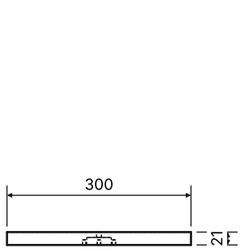
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

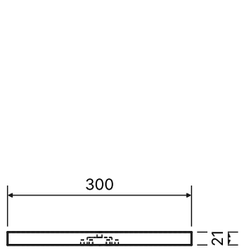
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

10. platter: lathe, cast-acrylic; 1467 cm3 each; 1 for the set; 1:10

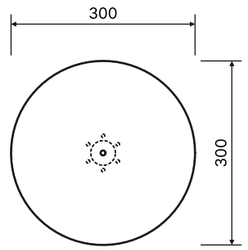
three laminae (tinted 3 / clear 12.7 / tinted 5.6) cut as 312 discs from the cover rings' offcuts, cemented and annealed, then turned: rough to 302, annealed again, both faces faced to 21.0, the rim finish-turned to 300 and polished, the clear stratum's rim crowned 0.2 for the belt; the bore 40.2 H7 for the hub's spigot; six blind 5.6 holes 8.5 deep on r 28 from below for brass M4 inserts (epoxy-bonded, no thread-locker near the acrylic): the screws come up through the hub, so no head bears on acrylic, and the record side is unbroken; the bond between the laminae carries no torque; flatness ...



FRONT



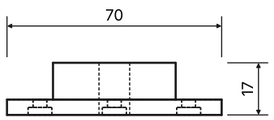
FROM THE RIGHT



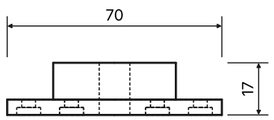
FROM ABOVE

11. platter-hub: lathe, brass-leadfree; 32 cm3 each; 1 for the set; 1:2

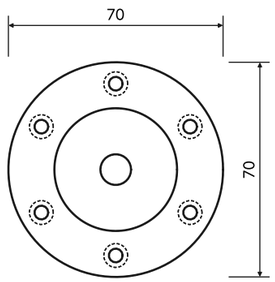
lead-free brass: a 70 × 5 flange, a 40 h6 spigot 12 long, six 4.5 holes on r 28 counterbored 7.8 × 2.4 from below for the screws' heads; shrunk onto the spindle (heated to 200 C) and finish-turned true to it after



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

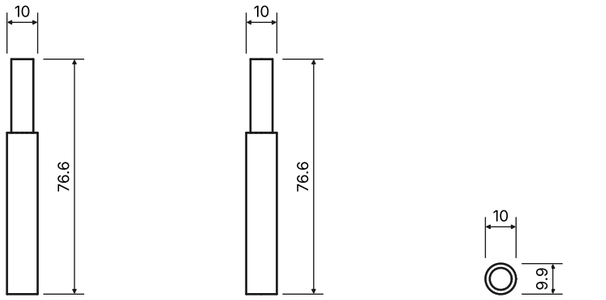
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

14. spindle: lathe, stainless-440c; 5 cm3 each; 1 for the set; 1:2

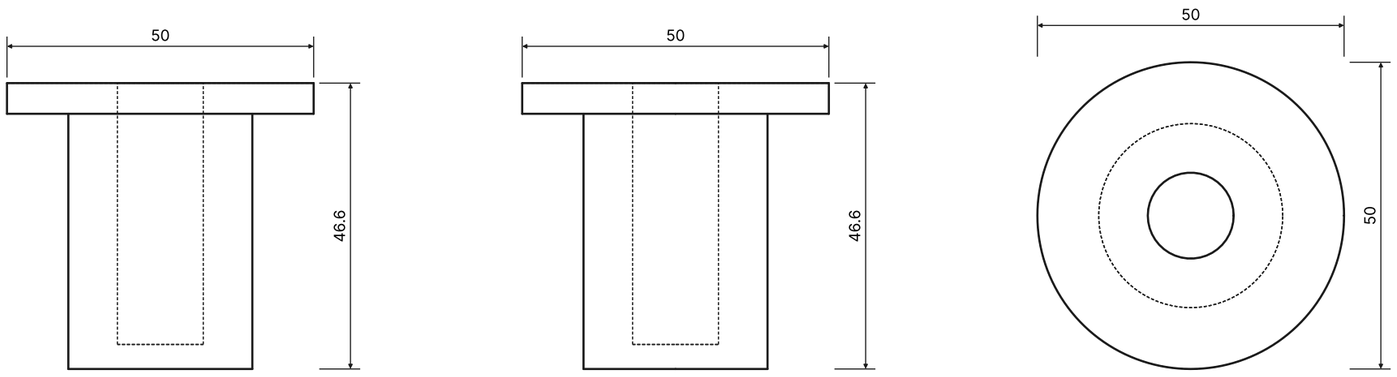
440C hardened to 58 HRC, ground to 10 g5 at the journals, Ra 0.1 (the drawing and the parts list now agree: not 304, which galls in bronze); the record spindle 7.2; a 6 cup at its foot for the ball; a vent flat ground along it, 1.0 wide x 0.3 deep over the journals' length, so air leaves the blind bore as the spindle goes in and the oil is not pumped out



FRONT **FROM THE RIGHT** **FROM ABOVE**

15. bearing-housing: lathe, brass-leadfree; 33 cm3 each; 1 for the set; 1:1

lead-free brass, threaded M30 × 1.5 for 30 from its foot, a 50 × 5 flange; the bushes pressed in and line-reamed or honed to 10 H6 (8-12 um running clearance on the g5 journal); clamped under the deck by its nut on one disc spring and a brass washer, set to about 1.5 kN by angle (no insert near the bore); 1.0 ml of oil (the space is about 1.2 ml), to about 10 under the upper bush



FRONT **FROM THE RIGHT** **FROM ABOVE**

Printed and machined parts

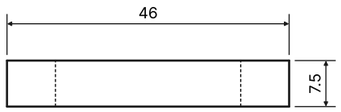
Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

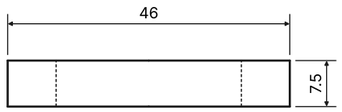
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

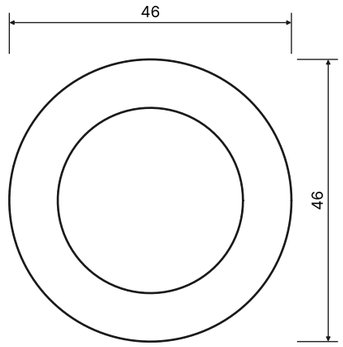
16. bearing-nut: lathe, brass-leadfree; 7 cm3 each; 1 for the set; 1:1



FRONT



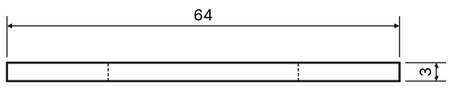
FROM THE RIGHT



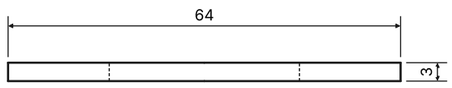
FROM ABOVE

18. bearing-washer: lathe, brass-leadfree; 7 cm3 each; 1 for the set; 1:1

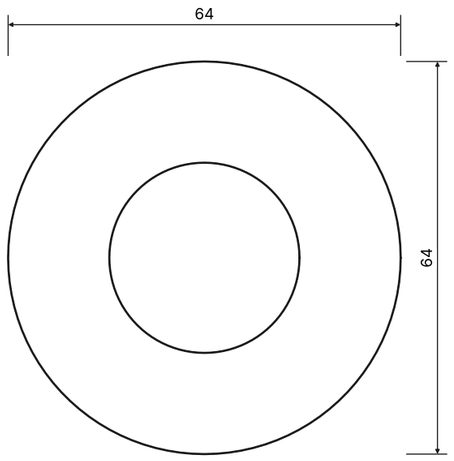
64 × 31 × 3, faced flat: spreads 1.5 kN over 2,460 mm2 of acrylic, 0.6 MPa



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

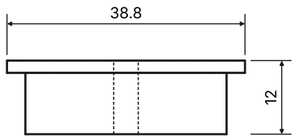
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

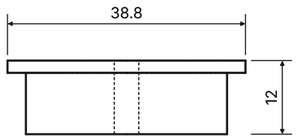
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

25. pulley: lathe, aluminium; 11 cm3 each; 1 for the set; 1:1

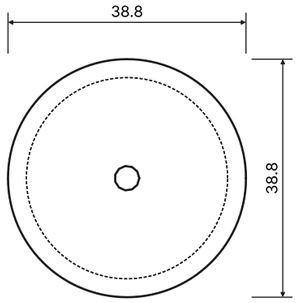
belt face 32.80 at 300 rpm for 33.3 (the Neo switches 45 electrically; trim 0.01 rpm); two M3 grub screws on a flat; PROPOSAL: not turned until the bench test (STOCK-AND-AUDIT.md 3.1) has measured the motor on the US Neo; a test pulley in aluminum first



FRONT



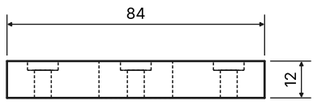
FROM THE RIGHT



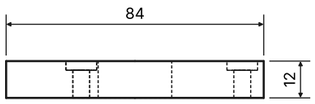
FROM ABOVE

27. arm-puck: lathe, aluminium; 60 cm3 each; 1 for the set; 1:2

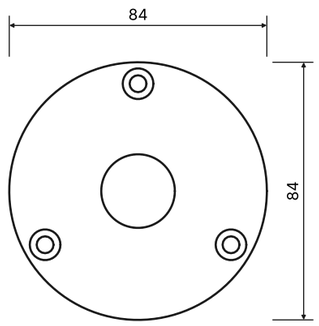
6061, 84 × 12, bead-blasted and clear anodised: Rega's 24 hole and three-point fixing (holes per Rega's template, UNCONFIRMED here); three M5 countersunk screws into brass inserts in the deck's shoulder. Another arm takes another puck



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

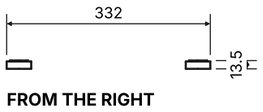
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

30. feet: lathe, stainless; 48 cm3 each; 1 for the set; 1:10

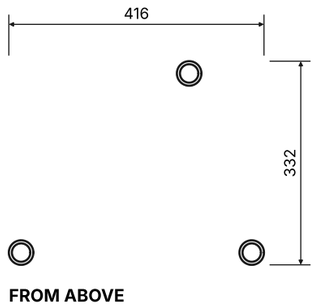
three, turned 303 stainless, bead-blasted, 40 × 12 with a 30 neck, a tapped M8 bore (brass read as amber through the tint: round 48's critics); each on a stud in the deck's M8 insert, a stainless lock nut over it: level the platter by turning them, then lock; they locate and level; no isolation is claimed (on pads this small they sit at 50 Hz or more, above the feedback band)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	deck-L1	cut	1	3 mm Cell-cast acrylic sheet, stock tint (TruCast 2064 Gray or 2423 Red), 4 × 8 ft	472 × 382	14.88
2	deck-L1-inlay	cut	1	3 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	432 × 342	6.65
3	deck-L2	cut	1	12.7 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	472 × 382	34.35
4	deck-L3	cut	1	3 mm Cell-cast acrylic sheet, stock tint (TruCast 2064 Gray or 2423 Red), 4 × 8 ft	466 × 376	14.46
5	deck-L3-inlay	cut	1	3 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	426 × 336	6.44
6	deck-L4	cut	1	12.7 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	466 × 376	33.38
7	deck-L5	cut	1	3 mm Cell-cast acrylic sheet, stock tint (TruCast 2064 Gray or 2423 Red), 4 × 8 ft	460 × 370	14.04
8	deck-L5-inlay	cut	1	3 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	420 × 330	6.24
9	deck-L6	cut	1	12.7 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	460 × 370	32.42
10	platter	lathe	1	Cast acrylic tube, tinted through	310 × 310 × 22	169.57
11	platter-hub	lathe	1	Brass C69300 (Eco Brass, lead-free) or C27450	80 × 80 × 27	70.92
12	platter-screws	buy	1	Six M4 × 10 stainless socket button screws (ISO 7380), up through the hub into the platter's inserts		6.00
13	platter-inserts	buy	1	any (E-Z Lok / Tappex kind) Six M4 brass threaded inserts for bonding into plastics, 5.6 OD x 8, epoxy-bonded into the platter from below (no thread-locker near acrylic)		9.00
14	spindle	lathe	1	Stainless steel 440C, hardened 58 HRC and ground	20 × 20 × 87	127.08
15	bearing-housing	lathe	1	Brass C69300 (Eco Brass, lead-free) or C27450	60 × 60 × 57	75.56
16	bearing-nut	lathe	1	Brass C69300 (Eco Brass, lead-free) or C27450	56 × 56 × 18	53.23
17	bearing-spring	buy	1	Schnorr / Mubea / Christianson kind One DIN 2093 group 2 series C disc spring 63 x 31 × 1.8 (51CrV4), under the bearing nut; load and rate to confirm in the maker's table		6.00
18	bearing-washer	lathe	1	Brass C69300 (Eco Brass, lead-free) or C27450	74 × 74 × 13	55.68
19	bushes	buy	1	any lathe shop Two solid bronze plain bushes (C932 / SAE 660), 10 × 14 × 12, pressed into the housing and honed or line-reamed to 10 H6 (8-12 um running clearance on a 10 g5 spindle)		40.00
20	ball	buy	1	Tungsten carbide ball, 6 mm, grade 10: the spindle's thrust point, seated in a cup at the spindle's foot		5.00
21	thrust-plate	buy	1	Thrust plate 13.8 × 3, hardened 52100 steel (60 HRC) lapped flat, at the foot of the housing's bore in the oil bath, under the carbide ball (a sapphire window of the same size is the alternative)		20.00
22	motor-plate	cut	1	3 mm Aluminium 5052 sheet	90 × 90	1.62
23	motor	buy	1	Rega 24 V low-vibration synchronous motor upgrade kit (motor, its board and mounting), for the Neo PSU		265.00
24	motor-screws	buy	1	Three M3 × 12 stainless button screws and three M3 brass inserts (4.6 OD), the motor plate hard to the pocket's shoulder as Rega fits its motor		6.00
25	pulley	lathe	1	Aluminium 6061	49 × 49 × 22	46.31
26	belt	buy	1	Flat turntable belt, ground rubber or polyurethane, 5 wide x 0.6, cut to the measured path (about 945 long for the 994 path, 5 % stretch); an O-ring test belt first		30.00

Parts list (1 of 3)

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
27	arm-puck	lathe	1	Aluminium 6061	94 × 94 × 22	49.86
28	arm	buy	1	Rega RB330 tonearm (9 in, Rega geometry: 222 mm from the platter's center to the arm's hole, effective length 236.5, overhang 14.5, offset 21.55 deg), with its own lead and cueing lift		675.00
29	cartridge	buy	1	Ortofon 2M Black moving-magnet cartridge (nude Shibata stylus), black body, 1/2 in mount		750.00
30	feet	lathe	1	Stainless steel 304	45 × 45 × 72	49.37
31	foot-pads	buy	1	Isolate It (Sorbothane Inc.) Three Sorbothane discs 40 × 4, 50 duro, bonded under the brass feet: about 45 N each, 36 kPa		20.00
32	foot-studs	buy	1	Three M8 × 30 stainless studs, three M8 brass lock nuts (thin), three M8 × 12 brass threaded inserts (for acrylic, epoxy-bonded)		18.00
33	phono-stage	buy	1	Pro-Ject Phono Box S2, MM/MC phono stage in an aluminum case, 103 × 115 × 36 (one listing 103 × 36 × 115 mm, 0.57 kg): RIAA within 0.4 dB 20 Hz-20 kHz, MM S/N 94 dBA at 40/43 dB gain, THD under 0.01 %, subsonic filter 20 Hz 12 dB/oct, MM 47k		269.00
34	phono-runners	buy	1	Four 10 mm felt pads under the phono stage		2.00
35	back-plate	cut	1	2 mm Lead-free brass sheet (C26000 / C27450), brushed	144 × 45	2.59
36	back-hardware	buy	1	Neutrik / Switchcraft / CUI kind Two gold panel RCA jacks (insulated, 1/4 in hole), a 4 mm earth binding post, a 4PDT signal toggle with gold contacts (C&K 7411 kind: in bypass it switches the phono stage's input and output, so the		62.00
37	cover-bumpers	buy	1	3M Bumpon SJ5302 clear, six (three under the cover, three spare)		4.00
38	dowels	buy	1	Two 5 × 12 stainless dowel pins, h6, pressed into the deck		4.00
39	inserts	buy	1	any (E-Z Lok / Tappex kind) Brass threaded inserts for plastics, epoxy-bonded: 3 M5 (the puck), 4 M3 (the back plate); with M5 × 12 countersunk and M3 × 6 button screws		14.00
40	wiring	buy	1	Shielded interconnects inside (the arm's lead to the S2, the S2 to the plate), a star earth at the plate's post (arm, bearing housing via an M3 ring tag, S2), two clear cord clips under the deck, a cord grip for the Neo's lead		25.00
41	neo-psu	buy	1	Rega Neo PSU MK2: DSP-generated drive for Rega's 24 V motor, 33/45 by a button, fine speed trim (0.01 rpm steps) on the back; US 60 Hz 115-120 V version		445.00
42	bearing-oil	buy	1	Turntable bearing oil, 8 ml (a light synthetic oil, ISO VG 32 or as the bush maker says)		15.00
43	cement	buy	1	Evonik (Röhm) ACRIFIX 2R 0190, two-component polymerising cement for cast acrylic (cures through any tint, unlike the one-part light-curing 1R 0192); Weld-On 40 or 42 is the US equivalent		45.00
44	cover-C1	cut	1	12.7 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	454 × 364	31.48
45	cover-C2	cut	1	3 mm Cell-cast acrylic sheet, stock tint (TruCast 2064 Gray or 2423 Red), 4 × 8 ft	454 × 364	13.63
46	cover-C3	cut	1	12.7 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	448 × 358	30.55
47	cover-C4	cut	1	3 mm Cell-cast acrylic sheet, stock tint (TruCast 2064 Gray or 2423 Red), 4 × 8 ft	448 × 358	13.23
48	cover-C5	cut	1	12.7 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	442 × 352	29.64
49	cover-C6	cut	1	3 mm Cell-cast acrylic sheet, stock tint (TruCast 2064 Gray or 2423 Red), 4 × 8 ft	442 × 352	12.84
50	cover-C7	cut	1	12.7 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	436 × 346	28.74

Parts list (2 of 3)

Mantle: Outcrop (proposal B): Belt-drive record player cut from a stack of bonded cast sheet, with a lift

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Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
51	cover-C8	cut	1	3 mm Cell-cast acrylic sheet, stock tint (TruCast 2064 Gray or 2423 Red), 4 × 8 ft	436 × 346	12.45
52	cover-C9	cut	1	12.7 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	430 × 340	27.85
53	cover-C10	cut	1	3 mm Cell-cast acrylic sheet, stock tint (TruCast 2064 Gray or 2423 Red), 4 × 8 ft	430 × 340	12.06
54	cover-top	cut	1	5.6 mm Cell-cast acrylic sheet, water-clear, 4 × 8 ft	424 × 334	11.90
55	cover-top-tint	cut	1	3 mm Cell-cast acrylic sheet, stock tint (TruCast 2064 Gray or 2423 Red), 4 × 8 ft	424 × 334	11.68
56	mark	buy	1	Laser etching into the cast acrylic, frosted, about 0.1 deep, from a vector file: EARF 44 wide, the etch filled with the color, deeper		15.00

Parts list (3 of 3)

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Notes

1. The deck: seven laminae of stock cell-cast sheet (clear 12.7 strata, tinted 3 contour lines), cemented in one press with ACRIFIX 2R 0190 under vacuum and annealed (80 C, 10 h, 5 C/h down); faces fly-cut, the terraced outline finish-routed, noses eased R1, outer faces diamond-polished. Holes and pockets satin, as machined.
2. The cover: five clear rings with a tinted lamina on each and a clear 5.6 top under a tinted 3, cemented and annealed as the deck; it lifts off on two stainless dowels, on three clear bumpers. Stood on its back edge it frames the sleeve playing.
3. The bearing: brass housing M30 × 1.5 clamped by its nut under the deck (10 N m on a PTFE washer); two C932 bronze bushes honed to 10 H6 on a 10 g5 440C spindle (8-12 um, a vent flat along it); a 6 carbide ball on a lapped 52100 plate; 1.0 ml ISO VG 32. The nut bears on one DIN 2093 disc spring and a brass washer, set to about 1.5 kN, so the preload holds over temperature. Shipping: platter off, the spindle in, a nitrile cap over the housing.
4. The platter: three laminae, turned, on a brass hub shrunk onto the spindle, six M4 screws up through the hub into bonded brass inserts, 0.6 N m, no thread-locker. No mat (acrylic is close to vinyl in acoustic impedance); a carbon-fibre brush and an earthed housing for static.
5. The motor: Rega 24 V on its plate, screwed hard to the pocket's shoulder as Rega fits it (no grommets); the Neo PSU MK2 beside the record player switches 33/45 and trims speed. The pulley is turned to the motor's speed as measured on the bench (300 rpm expected at 60 Hz).
6. The arm: Rega RB330 on an aluminum puck (Rega three-point) at 222 from the spindle; Rega spacers set the height to the record (28.8 over the deck's top, set on the first sample with the tube level).
7. The phono stage: Pro-Ject Phono Box S2 in a tunnel from the back, on felt runners; the brass plate closes it and carries line out, earth, a 4PDT bypass toggle and the 18 V inlet (Pro-Ject's own listed adapter). The arm's lead runs in a groove under the deck to it.
8. Feet: three, turned stainless on Sorbothane, on M8 studs into brass inserts; level the platter, then lock the nuts. They locate and level; no isolation is claimed. For isolation, a platform at 3-5 Hz under the record player.
9. Before the first sample: bench the motor on the US Neo (STOCK-AND-AUDIT.md 3.1); with the arm in hand, check its counterweight clears the cover's top rings.
10. First-sample checks: speed and wow (under 0.1 % weighted) with a test record; a tap test on the deck with the stylus on a still record; the bond lines checked against a light table for bubbles before the outline is finish-routed.

Notes

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